

1. Multiple Choice Questions (21%)

1) Which of the following statements correctly identifies the Second Law of Thermodynamics?

- A) All spontaneous processes increase the entropy of the universe; B) The enthalpy of a pure substance at 0 K is zero; C) The total entropy of the universe is the sum of the entropy of the system and the surroundings; D) The free energy of a reaction at equilibrium is zero; E) The reverse of a spontaneous reaction is a nonspontaneous process.

2) What is the difference between extensive and intensive properties?

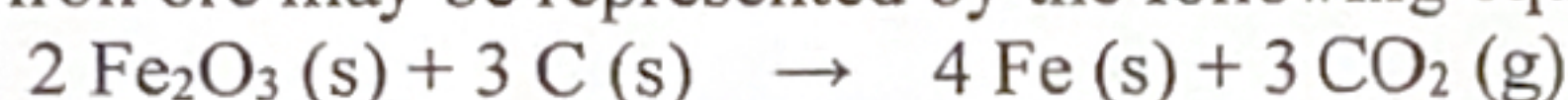
- A) Extensive properties depend on the quantity of matter, while intensive properties do not. B) Extensive properties occur outside a system, while intensive properties occur inside. C) Extensive properties are state functions, while intensive properties are not. D) Extensive properties include color and smell, while intensive properties include enthalpy and mass. E) Extensive properties depend on the kinetic energy of a system, while intensive properties depend on the potential energy.

3) Which of the following substances have a standard enthalpy of formation of 0?

- A) Hg(l); B) C(graphene); C) Rb(g); D) O₂(l).

4) What is the ΔS for the heating of 2 mol of water from 10°C to 60°C, given the heat capacity (C_p) of water is 4.18 J/g°C? A) 24.5; B) 0.337; C) 6.06; D) 2.90 J/K.

5) The extraction of iron from iron ore may be represented by the following equation:

Given the information below, calculate ΔH° for the process per mole of Fe₂O₃

$$\Delta H_f [\text{Fe}_2\text{O}_3 (\text{s})] = -826 \text{ kJ mol}^{-1}; \Delta H_f [\text{CO}_2 (\text{g})] = -394 \text{ kJ mol}^{-1}$$

- A) -470 kJ; B) -2834 kJ; C) -1220 kJ; D) +235 kJ; E) +470 kJ

6) For which of the following processes would the entropy change be positive?

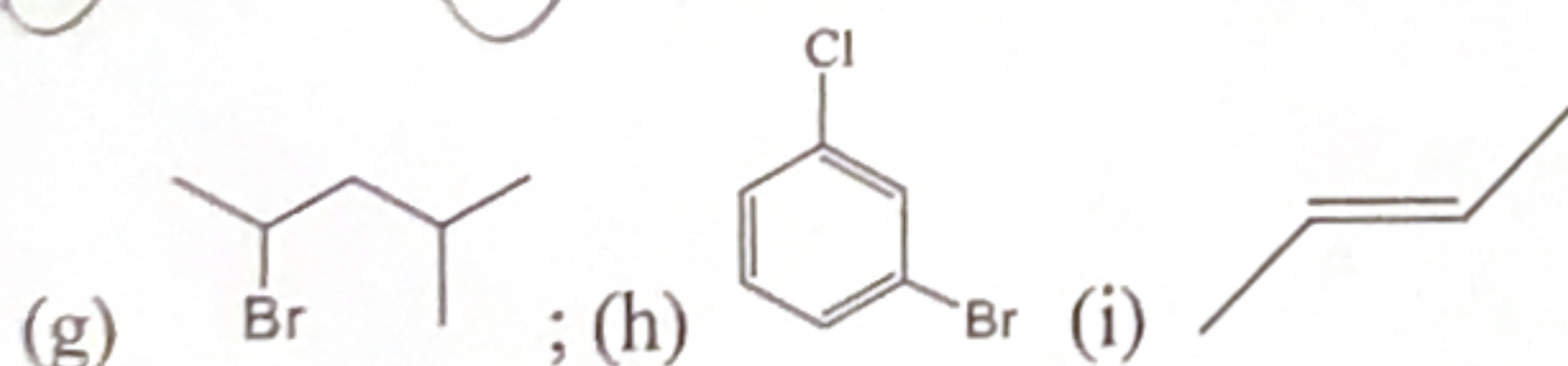
- A) Hg (l) (45°C) → Hg (l) (25°C); B) H₂O (l) → H₂O (s); C) He (g) (1 atm) → He (g) (10 atm); D) I₂ (s) → I₂ (aq); E) Ag⁺ (aq) + Br⁻ (aq) → AgBr (s).

7) When gasoline burns in a car engine, the heat released causes the products of CO₂ and H₂O to expand. Excess heat is removed by the car's cooling system. If the expanding gases do 4.51 kJ of work on the pistons and the system loses 511 J to the surroundings as heat, what is the change in internal energy?

- A) +0.96; B) -0.60; C) -960; D) -0.96 kJ.

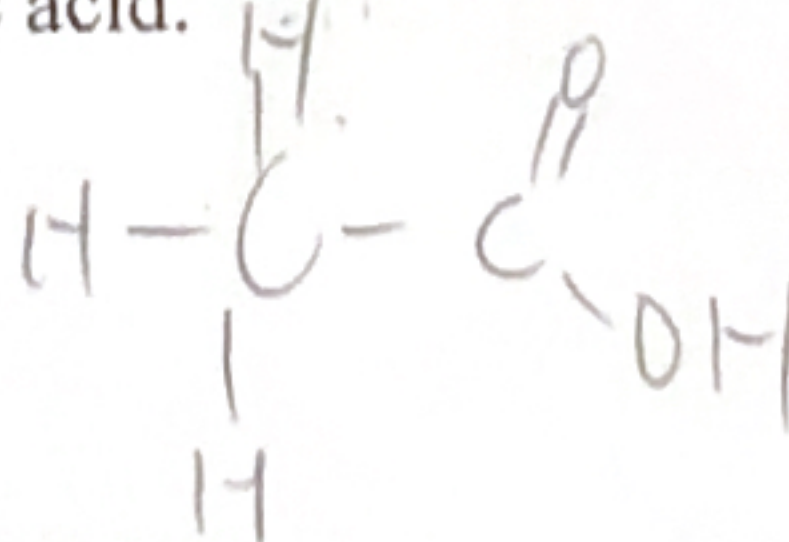
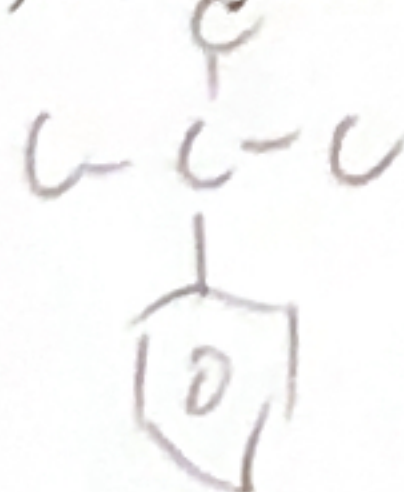
2. Name the following compounds, (27%)

- (a) NaHCO₃; (b) Pb(NO₃)₂; (c) SiH₄; (d) CH₂OHCHOHCH₃; (e) CN⁻; (f) NaClO₃;



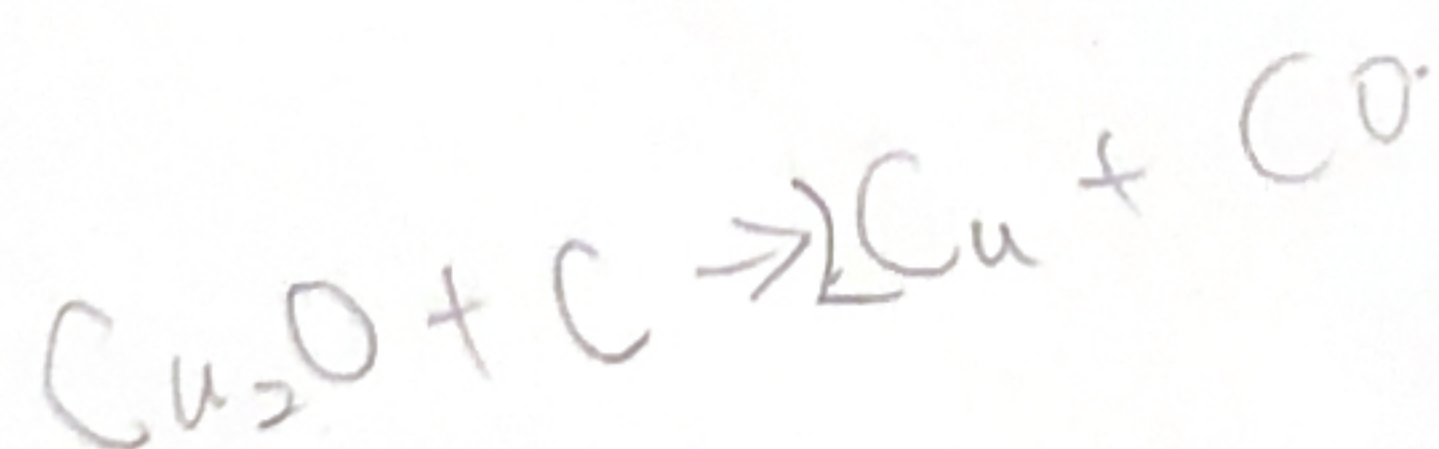
3. Draw the chemical structure of the following compounds, (9%)

- (a) *t*-butylbenzene; (b) acetylene; (c) acetic acid.



4. Zinc and HCl react according to the reaction, $\text{Zn}_{(s)} + 2\text{HCl}_{(aq)} \rightarrow \text{Zn}^{2+}_{(aq)} + 2\text{Cl}^{-}_{(aq)} + \text{H}_{2(g)}$. When 6.00g of Zinc metal is added to a dilute HCl solution at 1.00 atm and 25°C, and the above reaction is allowed to go to completion at constant pressure, 13.98 kJ of heat must be removed to the final solution to its original temperature. What are the values of work and the change of internal energy? (Hint: MW = 65.409 g mol⁻¹ for Zn). (6%)
5. (a) Calculate the change in both enthalpy and entropy when a 3.0g block of ice melts at 0.0°C [$\Delta H^{\circ}_{\text{fus}}(\text{H}_2\text{O}) = 6.01 \text{ kJ/mol}$]. (b) For the same block of ice, calculate the entropy change for the system when the ice is warmed from 0.0°C to 25°C. The heat capacity of liquid water over this temperature range is 75.3 J/mol°C. (8%) $C_p = 1$
6. Calculate (a) ΔG° and (b) $\Delta G^{\circ}_{300^{\circ}\text{C}}$ for the reaction $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$, assuming that ΔH and ΔS do not change between 25°C and 300°C. (c) Is this reaction spontaneous at 300°C? (d) Do you think we can increase the temperature and apply this reaction for industrial products from thermodynamic point of view? Explain your answer. (Hint: $S^{\circ}(\text{N}_2) = 191.6 \text{ J/(mol}\cdot\text{K)}$, $S^{\circ}(\text{H}_2) = 130.7 \text{ J/(mol}\cdot\text{K)}$, $S^{\circ}(\text{NH}_3) = 192.8 \text{ J/(mol}\cdot\text{K)}$, and $\Delta H^{\circ}_f(\text{NH}_3) = -45.9 \text{ kJ/mol}$) (13%)
7. Calculate the equilibrium constant at 25°C for each equilibrium reaction,
 (a) $\text{CCl}_{4(g)} + 6\text{H}_2\text{O}_{(l)} \rightleftharpoons \text{CO}_{2(g)} + 4\text{HCl}_{(aq)}$; $\Delta G^{\circ} = -377 \text{ kJ/mol}$
 (b) $\text{PCl}_{3(g)} + \text{S}_{(s)} \rightleftharpoons \text{PSCl}_{3(l)}$; $\Delta G^{\circ}_f: \text{PCl}_3 = -272.4 \text{ kJ/mol}$, $\text{PSCl}_3 = -363.2 \text{ kJ/mol}$
 (Hint: ΔG°_f is the standard free energy of formation) (6%)
8. Assume that a particular copper ore consists of pure cuprous oxide (Cu_2O). Using ΔG°_f values given, calculate (a) ΔG° and K_p for the decomposition of cuprous oxide to metallic copper and oxygen gas [$\text{Cu}_2\text{O}_{(s)} \rightleftharpoons 2\text{Cu}_{(g)} + 1/2\text{O}_{2(g)}$] and (b) ΔG° and K_p for the reaction of cuprous oxide with carbon to produce metallic copper and carbon monoxide. (Hint: $\Delta G^{\circ}_f(\text{Cu}_2\text{O}) = -146.0 \text{ kJ/mol}$; $\Delta G^{\circ}_f(\text{CO}) = -137.2 \text{ kJ/mol}$) (10%)

$$178.314 \times 298 \ln \left(\frac{6^{2.09}}{1} \right)$$



$$\frac{5}{2} \times 0.082 \times 298 + 1$$